

A1 · Cover and control identity

Enterprise Compliance Master Record for case steel_in_fixture. This document is the operator's permanent corporate record; it is not a copy of the verifier dossier and it never implies an independent verification opinion.

Control	
Document	
Operator	
Installation	
Reporting period	
Package code	
Release	
Generated (UTC)	
Schema / engine	
Ruleset	
Single authoritative state	
Retained until	
Confidentiality	

DATA & EVIDENCE READINESS

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PERIOD CLOSURE

100%

Calendar-only; reflects elapsed period days

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SINGLE AUTHORITATIVE STATE

READY_FOR_INDEPENDENT_VERIFICATION — PERIOD_CLOSED · NO_OPERATOR_CLOSABLE_FINDINGS. An open repo negative label.

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READY_FOR_INDEPENDENT_VERIFICATION — PERIOD_CLOSED · NO_OPERATOR_CLOSABLE_FINDINGS. An open report with no negative label.

A3 · What this document is and is not

This document does

- Record the full engagement: scope, results, rationale and evidence chain.
- Explain how every sealed number was produced and how to reproduce it.
- Provide a retention and integrity-verification guide for the legal retention period.

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Explain how every sealed number was produced and how to reproduce it.

Provide a retention and integrity-verification guide for the legal retention period.

Give the operator a plain-language action list grouped by owner.

This document does not

- Express an independent verification opinion or a conformity conclusion.
- Replace a binding customs classification decision or any customs ruling.
- Submit anything to the CBAM Registry on behalf of the operator.

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Express an independent verification opinion or a conformity conclusion.

Replace a binding customs classification decision or any customs ruling.

Submit anything to the CBAM Registry on behalf of the operator.

Guarantee acceptance, approval or compliance of the
 installation record; the CBAMValid system is the

automated preparation, sealing and reproduction tool; the accredited independent verifier performs the verification work and records its own opinion. The importer remains responsible for any Registry submission derived from this material.

A5 · Reading guide

Question	
What did this package produce?	
Is my data ready for independent verification?	
Which evidence gaps must I close?	
What does my registry field mapping look like?	
How is my emissions number calculated?	
Can a third party reproduce the numbers?	
Which hashes cover what?	
What is in and out of the system boundary?	
Which methodology decisions were made and rejected?	
What happens if my period is still open?	
How long must I keep this package and how do I verify integrity?	
What is next for the following period?	

B1 · Delivery inventory

Component			
CBAMValid Verification Readiness & Evidence Assurance Dossier.pdf			
Complete Dossier Compilation.pdf			
Product Scope Assessment.pdf			
CN Code Reasoning.pdf			
Required Data Checklist.pdf			
Installation Monitoring Plan.pdf			
Production Process Map.pdf			
System Boundary Register.pdf			
Source Stream Register.csv			
Emission Source Register.csv			
Measurement and Meter Register.csv			
Activity Data Ledger.csv			
Evidence Register.csv			
Field-to-Evidence Matrix.csv			
Methodology Decision Log.pdf			
Embedded Emissions Calculation Annex.pdf			
Operator Emissions Report.pdf			
Misstatement and Non-Conformity Register.csv			
Corrective Action Log.csv			
O3CI Field Mapping.csv			
Calculation Trace.json			

B3 · Out of scope and limits

Not covered by this package	
Independent verification opinion	
Binding customs tariff information	
CBAM Registry submission	
Site visit and physical inspection	
Materiality judgement	

C1 · Two-axis readiness panel

DATA & EVIDENCE READINESS

100/100

Everything under your control. Loss reasons: none.

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PERIOD CLOSURE

100%

Everything the calendar controls. The data score never includes the calendar.

100%

Everything the calendar controls. The data score never includes the calendar.

The two axes are intentionally independent. Buying before the reporting period closes never stamps your package as failed; it reports an honest ON_TRACK state while your data is complete.

C2 · Score breakdown

Component					
Identity and installation					
Scope and methodology					
Activity data					
Evidence completeness					
Calculation integrity					
Allocation and reconciliation					
Data quality and uncertainty					
Package integrity					

Every lost point is explained: each component shows the score and loss reason from the sealed assessment, and integrity penalties below explain any further reduction of the data & evidence readiness axis.

C3 · Findings register

Finding ID					
NONE					

Every registry field declared complete without mandatory evidence automatically produced a finding above. Findings are never suppressed.

C4 · Action list

Owner

No technical terms above: each row states who does what, by when and with which document.

C5 · Preparation for the next sealed version

A new sealed version is required when input values, allocation shares, evidence records or methodology decisions change. Start collecting the following evidence for the next period now: metered consumption records, updated grid factor certificates, customs declarations, and calibration certificates before their validity expires.

Any change to a sealed value invalidates the previous package hash chain; the replacement package carries a new report identity and supersedes the earlier release.

D1 · Parties

Party attribute		
Importer		
EORI		
Operator		
Installation		
Country		
Production route		

D2 · Installation and production route

Installation Jamshedpur Integrated Steel Plant; production route Blast Furnace Route (BF-BOF). The sealed package covers the definitive reporting period 2026-ANNUAL.

SYSTEM BOUNDARY
Boundary includes coke preparation, sinter plant, blast furnace, basic oxygen furnace, casting and finishing operations. Excludes downstream rolling mills and on-site power export.

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D3 · System boundary

IN SCOPE Direct emissions, electricity, precursors, covered goods
--

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--

EXCLUDED Downstream rolling mills and exported by-product power
--

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--

Every exclusion requires a legal or technical basis in the monitoring plan. The boundary is operator-declared; the independent verifier confirms conformity with the sector-specific legal boundary.

D4 · Goods population

Good						
1						
2						
3						

D5 · CN classification rationale

Good				
1				
2				
3				

CUSTOMS DECISION BOUNDARY

This record documents the classification supplied in the sealed case. It does not replace a binding customs classification decision; any disputed CN code remains subject to customs review.

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E2 · Emission source register

Source						
Direct emissions						
Electricity						
Grid factor						

E3 · Measurement and meter register

Input				
Direct emissions				
Electricity				
Grid factor				

E4 · Evidence register

[illegible]

SHA-256 prefixes are printed above; full hashes are stored in Evidence Register.csv and Data Integrity Manifest.json inside the package.

E5 · Grading methodology

Grades A-E assess the source and integrity of each document: A official/regulatory source with strong provenance; B accredited or certified authority; C reputable institutional source; D secondary or self-published material; E unstructured or low-provenance material.

WHAT A GRADE DOES NOT PROVE

A high grade is an assessment of a document's source and integrity; it is not an independent verification result about the accuracy of its content.

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Independent verifiability is a separate axis: an APPROVED and malware-clean record with a stable hash is independently checkable; a PENDING record is not.

F1 · Calculation summary

Category			
A — Installation direct			
B — Precursor direct			
C — Total direct (A+B)			
D — Electricity indirect			
E — Precursor indirect			
F — Total indirect (D+E)			
G — Certificate-relevant embedded			
H — Total informational embedded			

F3 · Calculation graph

CBAM_INSTALLATION_DIRECT_EMISSIONS
874300 tCO₂e · #b1217cfc

CBAM_INSTALLATION_DIRECT_EMISSIONS
874300 tCO₂e · #b1217cfc

CBAM_INDIRECT_EMISSIONS

388685 tCO₂e · #29513f98

CBAM_INDIRECT_EMISSIONS

388685 tCO₂e · #29513f98

CBAM_PRECURSOR_EMISSIONS_SUM
86400 tCO2e · #f702809a

CBAM_PRECURSOR_EMISSIONS_SUM
86400 tCO2e · #f702809a

CBAM_TOTAL_EMBEDDED_EMISSIONS_DISCLOSED
1349385 tCO2e · #e68d59cf

CBAM_TOTAL_EMBEDDED_EMISSIONS_DISCLOSED
1349385 tCO2e · #e68d59cf

CBAM_GOOD_EMISSIONS_ALLOCATION_1

CBAM_GOOD_EMISSIONS_ALLOCATION_1

CBAM_GOOD_EMISSIONS_ALLOCATION_2 2.106007 tCO2e/t · #3c207261

CBAM_GOOD_EMISSIONS_ALLOCATION_2 2.106007 tCO2e/t · #3c207261
--

CBAM_GOOD_EMISSIONS_ALLOCATION_3
2.113556 tCO₂e/t · #3f07de20

CBAM_GOOD_EMISSIONS_ALLOCATION_3
2.113556 tCO₂e/t · #3f07de20

CBAM_TOTAL_EMBEDDED_EMISSIONS_PRICED 951100 tCO₂e · #f2552465

CBAM_TOTAL_EMBEDDED_EMISSIONS_PRICED 951100 tCO₂e · #f2552465

CBAM_GOODS_ALLOCATION_RECONCILIATION
0 tCO2e · #ecb22e7b

CBAM_GOODS_ALLOCATION_RECONCILIATION
0 tCO2e · #ecb22e7b

CBAM_AGGREGATE_SPECIFIC_EMBEDDED_EMISSIONS

2.113556 tCO2e/t · #f2efb891

CBAM_AGGREGATE_SPECIFIC_EMBEDDED_EMISSIONS

2.113556 tCO2e/t · #f2efb891

CBAM CARBON PRICE REDUCTION RECORD TOTAL

150000 certificate-equivalent · #cc2ea44d

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The graph is a dependency DAG. Every node carries value, unit and hash prefix; edges show data flow. The root hash seals the whole graph.

F4 · Recalculation instruction

- 1 Collect the input values and their sources from E1-E3: direct emissions (tCO₂e), electricity consumption (MWh), grid emission factor (tCO₂e/MWh), precursors and goods production (t).
- 2 Verify unit conversion: kilograms to tonnes are divided by 1000; all other registered units are identity.
- 3 Apply the closed-form arithmetic for every trace node. Example: 280000 MWh x 0.4344 tCO₂e/MWh = 121632 tCO₂e.
- 4 Apply the rounding rule once, at presentation: Decimal.js precision 34, half-up to six decimals for per-good specific emissions only. No intermediate binary floating-point arithmetic.
- 5 Reproduce every hash: canonical serialisation — UTF-8, no BOM, object keys sorted lexicographically, JSON string escaping, decimal numbers without exponent, arrays in declared order, no whitespace between tokens — then SHA-256 over the UTF-8 bytes (HASH_CANONICAL_RULE 1.0).
- 6 Run the offline verifier: node Supporting_Evidence/verify/cli.js --package <extracted-zip-root> --strict. Expected output: manifest hash matches, detached signature verifies, ZIP byte hashes match, calculation root matches.

Hash			
manifestHash			
calculationRootHash			
calculationGraphRootHash			
legalSourceRegistryHash			

HASH CONSISTENCY

All formula hashes are identical under the same name across Calculation Trace and Calculation Graph.

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Canonical serialisation rule: Encoding: UTF-8, no BOM. Field order: object keys sorted lexicographically (localeCompare, byte order). Strings: JSON.stringify escaping (double quotes). Numbers: canonical decimal representation; integers without exponent. Arrays: elements in declared order. null/undefined/missing: serialised as the literal null. Whitespace: none between tokens; empty object {} and empty array []. Digest: SHA-256 over the canonical UTF-8 bytes, lowercase hexadecimal.

F5 · Allocation and reconciliation

Control	
Allocation method	
Allocation share total	
Allocation reconciliation delta	
Eligible certificate reduction	

A non-zero reconciliation delta is only acceptable with a written explanation; the sealed package records the delta value above.

G1 · Methodology decisions

Topic				
GOODS_EMISSIONS_ALLOCATION				
SYSTEM_BOUNDARY				

Every decision lists its rejected alternative and the reason it was rejected. A decision without an alternative is an incomplete methodology decision and produces a finding.

G3 · Registry field mapping

Field ID					
REG-OP-LEGAL-NAME					
REG-OP-REG-NO					
REG-OP-ADDRESS					
REG-OP-COUNTRY					
REG-OP-CONTACT-EMAIL					
REG-INST-NAME					
REG-INST-REGISTRY-ID					
REG-INST-UNLOCODE					
REG-INST-ADDRESS					
REG-INST-LATITUDE					
REG-INST-LONGITUDE					
REG-INST-COUNTRY					
REG-INST-PRODUCT-ROUTE					
REG-INST-SYSTEM-BOUNDARY					
REG-INST-EXCLUDED-PROCESSES					
REG-INST-MONITORING-PLAN					
REG-PERIOD-YEAR					
REG-PERIOD-START					
REG-PERIOD-END					
REG-GOOD-CN-0					
REG-GOOD-PROD-0					
REG-GOOD-ALLOC-0					

G4 · Risk register and sampling

[illegible]

No sampling population is left NOT_ASSESSED in this package. Any population deliberately not sampled is recorded as ASSESSED_NOT_SAMPLED_WITH_BASIS with its rationale in the sampling plan.

G5 · Responsibility matrix (RACI)

Work item			
Provide input data and evidence			
Assess evidence and grade			
Compute embedded emissions			
Record methodology decisions			
Map registry fields			
Close evidence gaps			
Verify the package			
Issue verification opinion			

H1 · Operator declaration and signatures

Role			
OPERATOR_PREPARER			
INTERNAL_REVIEWER			
DATA_OWNER			

DECLARATION

The operator declares that the recorded data reflects its records and that the evidence chain is complete to the best of its knowledge. This declaration is an operator record; it is not a verification opinion.

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H2 · Compliance calendar

Milestone	Q1 2024	Q2 2024	Q3 2024	Q4 2024
Reporting period close				
Data and evidence readiness				
Evidence gap closure				
Independent verification				
Registry submission (declarant)				

PERIOD CLOSE

The definitive reporting period 2026-01-01 to 2026-12-31 has 0 day(s) remaining and the calendar axis never modifies the data readiness score.

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H3 · Version history

Version				
V1				

H4 · Retention and integrity verification guide

Keep the whole package unmodified until 2031-01-31. Legal retention covers the operator's obligation to produce evidence for its CBAM declarations. Never split the package; keep the ZIP, manifest and signature together.

- 1 Open the package and verify the manifest hash: the manifest value for Data Integrity Manifest.json must equal the value shown on this page (A2).
- 2 Verify the detached signature: `node Supporting_Evidence/verify/cli.js --package <path> --strict` reports signature verification.
- 3 Compare file hashes: each component's SHA-256 in the manifest must equal the hash of the stored file.
- 4 If verification fails, the package has been altered or corrupted. Do not use it as evidence; request a fresh sealed release from the operator's records.

Public key reference: the general public key and key version are published by the operator's records; the signature record inside the package names the algorithm and key version used.